

Induction of Decision Trees:
”ICELL Tree Induction Software”

Olcay Taner Yıldız

Technical Report, FBE/COE-03/2004-14

Institute of Graduate Studies in Science and Engineering

Department of Computer Engineering

Bogaziçi University

yildizol@cmpe.boun.edu.tr

August 31, 2004

Acknowledgements

I would like to thank Prof. Ethem Alpaydn for his support and advises.

TABLE OF CONTENTS

Acknowledgements	1
1. Introduction	3
1.1. Installation	3
1.2. Generic Dataset Definition	4
2. C4.5	7
3. Nearest Mean Classifier	18
4. Classification and Regression Trees	20
5. Linear Discriminant Trees & QUEST	22
6. ID-LP & ID-MLP & Omnivariate Decision Tree	28
7. Regression with Multilayer Perceptron	31
8. LMDT	33
9. OC1	34
10. Options	35
10.1. Tree Options	35
10.2. View Options (Viewing train, test and prune data after run)	36
11. Batch Runs	37
REFERENCES	50

1. Introduction

ICELL Tree Induction software is programmed by Olcay Taner Yıldız in Delphi Programming Language. Current version of ICELL Tree induction software is 1.3.738. The software package contains

- Several decision tree algorithms such as C4.5 [7], CART [1], LDT [8], QUEST [5], OC1 [6], ID-LP [9], ID-MLP [4], LMDT [2] and ODT [9].
- Simple machine learning algorithms such as Nearest Mean, Multilayer Perceptron [3].
- Generic dataset definition interface, where you can define datasets by giving the general properties of the dataset and the data file of the dataset.
- User interface to see the training, testing and validation data.
- Batch run interface, where you can run several algorithms on several datasets.
- Adobe Acrobat interface, where you can view any pdf file from the software package.
- Online help.

1.1. Installation

To install ICELL Tree Induction Software follow these steps:

1. Click setup.exe to run the setup program.
2. Click Next.
3. Select the destination directory using Browse button and click Next.
4. Select the setup type from the possible options of Typical, Compact and Custom.
In compact option, only the files for the program to run are installed. In typical option, all files are installed. In custom option, you can select the files to install. Click Next.
5. Select the program folder to install. Click Next.

ICELL Tree Induction Software also requires Adobe Acrobat Reader to run. Therefore if Adobe Reader is not installed in your computer, you must install it before running the ICELL software. Any version of Adobe Reader is enough.

1.2. Generic Dataset Definition

Before running any algorithm on any dataset, you must define that dataset in ICELL. This is done under the Data Definition tab. All datasets in ICELL must be put in subdirectories of a common data directory. This data directory could be set in Directories options under Options tab. See Figure 1.1.

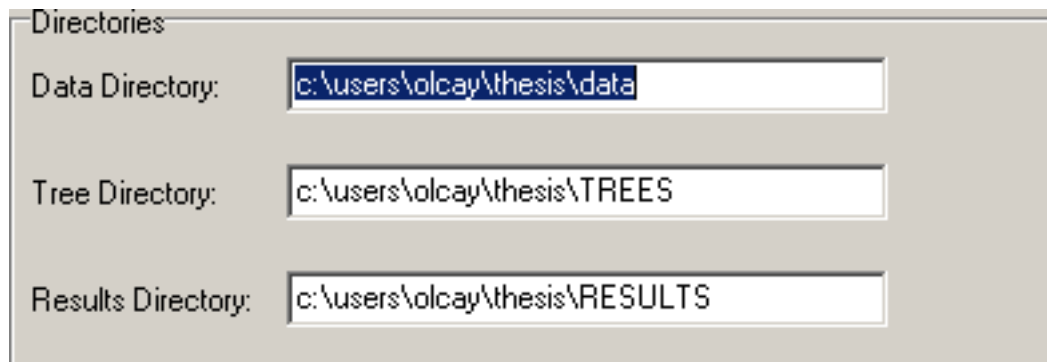


Figure 1.1. Data Directory selection in ICELL

In order to define a dataset:

1. Press New Data button.
2. Enter the name of the dataset in the dialog box that appears. After that the name of the dataset will appear as selected in Name combobox.
3. Enter the separator character in Separator edit box. You may use any separator character except tab character in your dataset. If your separator character is space put a space there.
4. Define the features that occurs in the data one after another by
 - First select the feature type from the list box. The feature type may be:
 - ClassDef for class definition attribute

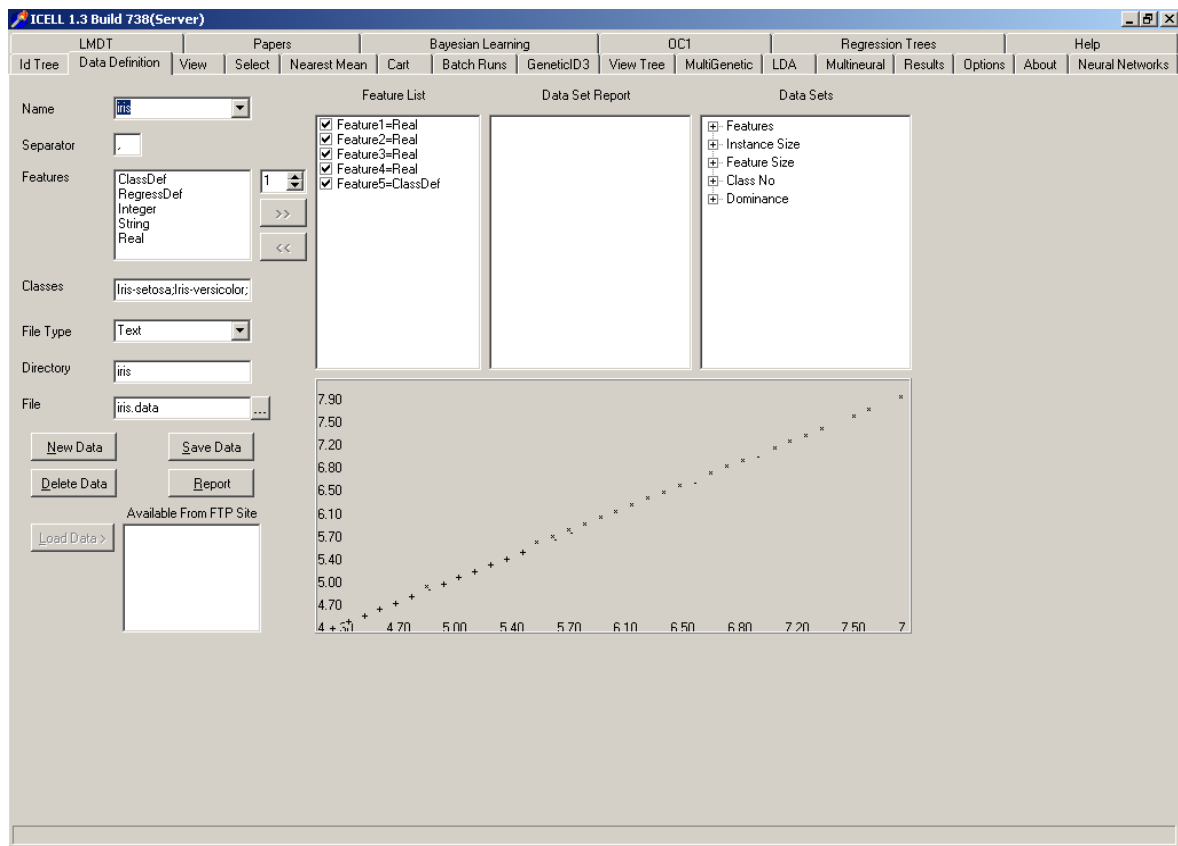


Figure 1.2. Dataset Definition Tab in ICELL

- RegressDef for the attribute that will be estimated using other attributes
 - Integer for integer valued attributes
 - Real for real valued attributes
 - String for discrete valued attributes
 - Select the number of attributes those have that selected feature type. By using this number you can define any number of attributes at once.
 - Press >> button.
 - Some of the attributes in the dataset may not be used in induction of algorithms. To make a feature available press the checkbox that occurs before its name. In order to select all of the features once, right click on the Feature List list box and press Select All menu item in the Popupmenu.
5. Enter classes separated by ; (semicolon) character in classes edit box.
 6. Select file type from File Type combobox. Currently only text files are supported.

7. Enter the subdirectory for this dataset under the Dataset directory in the Directory edit box.
8. Enter the file name of the data in the File edit box. You may also select the data file via a Windows File Dialog box by pressing ... button.
9. Press Save Data button.

After defining the datasets, you are ready to run tree induction and other algorithms on these datasets.

2. C4.5

You can run C4.5 algorithm from ICELL in two ways:

- ICELL will randomly divide your dataset into two parts (train set, test set), train C4.5 algorithm on the train set and test the resulting tree on the test set. To accomplish this:
 - Select one of the possible datasets from the dataset combobox (Figure 2.1).

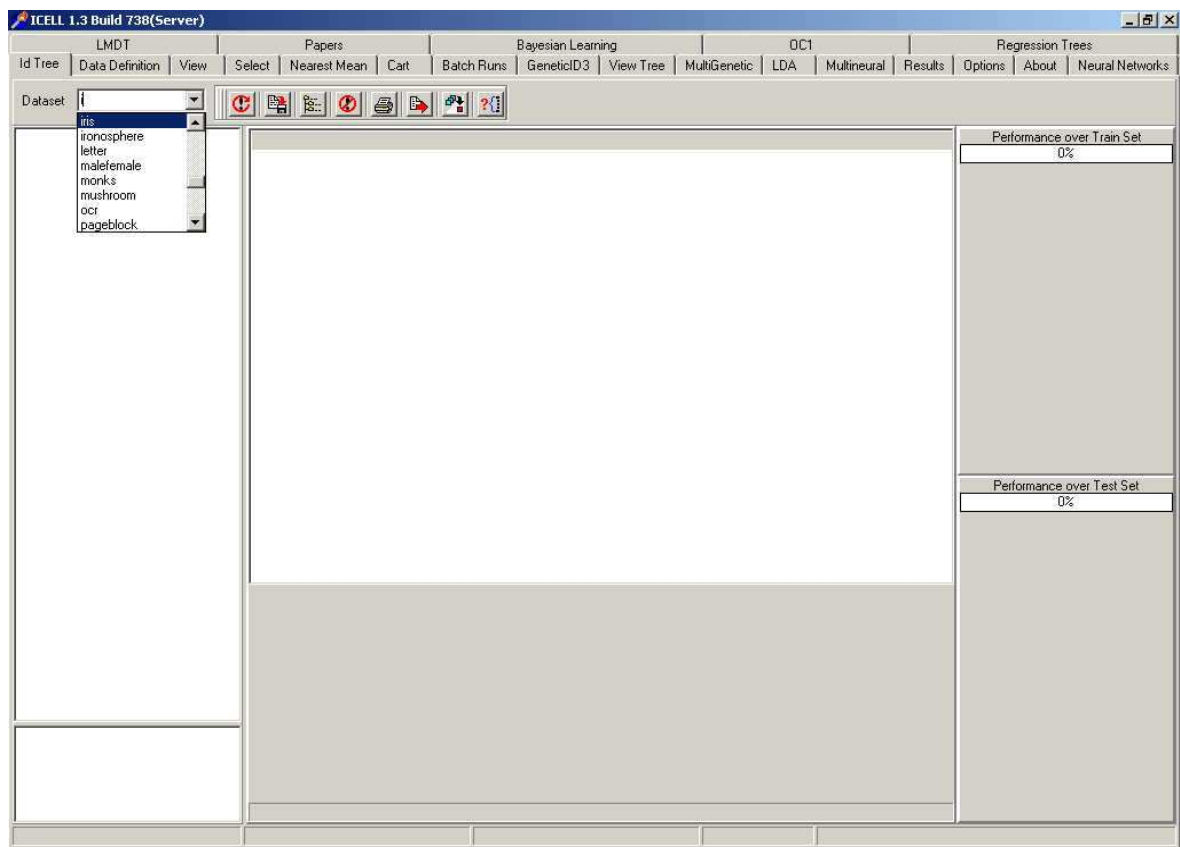


Figure 2.1. Selection of a dataset for C4.5 algorithm in ICELL

After running the algorithm you can save train and test files by pressing second button on the IdTree tab.

- You will select the train and test sets and ICELL will run C4.5 algorithm again on your selected train set and test the resulting tree on your test set. To accomplish

this:

- Select one of the possible datasets from the dataset combobox on the select tab (Figure 2.2).

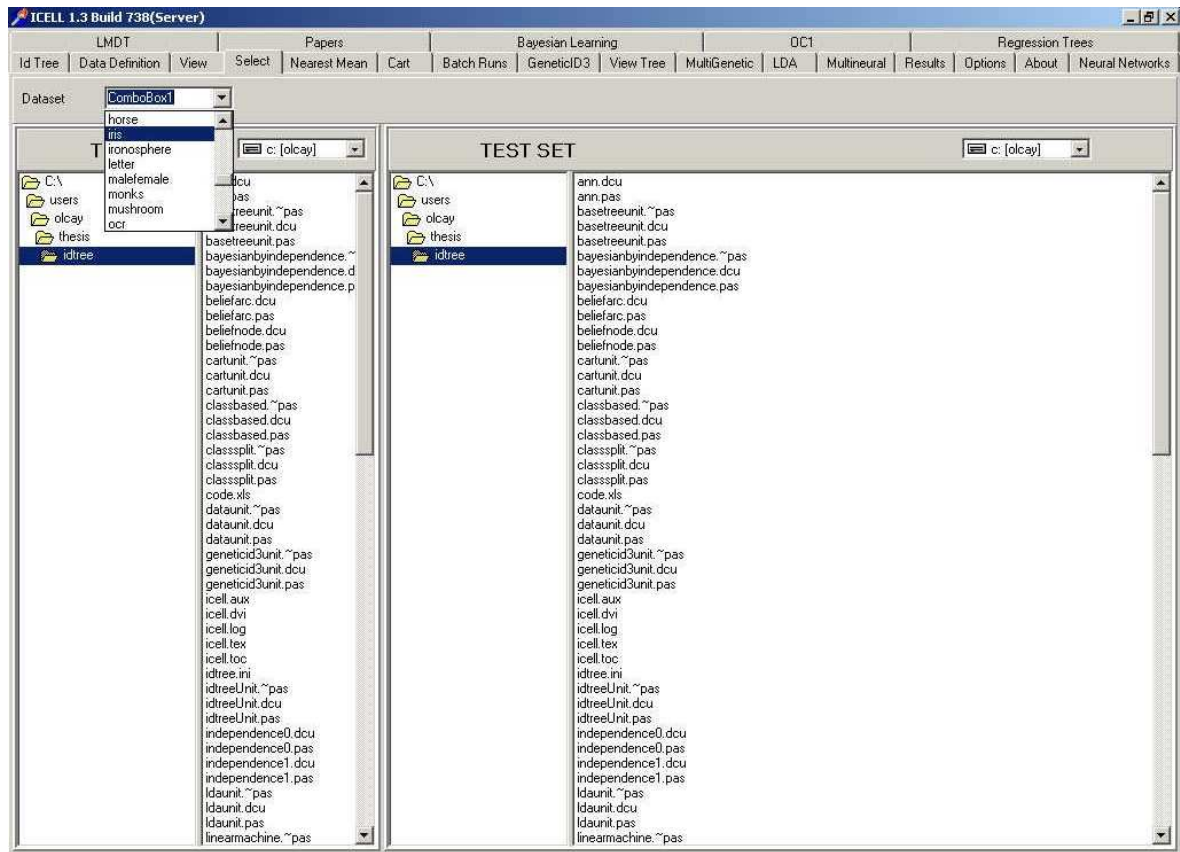


Figure 2.2. Selection of a dataset for C4.5 algorithm in ICELL

- Select training set from the train set panel, and test set from the test set panel (Figure 2.3).
- Press third button on the IdTree tab.

After running the algorithm, you may validate the algorithm using the test set as train set and train set as test set. For this press first button on the Idtree tab.

While running C4.5 algorithm you may:

- View current performances of the algorithm on the train and test sets (Figure 2.4).
- Stop running the algorithm by pressing fourth button on the IdTree tab.

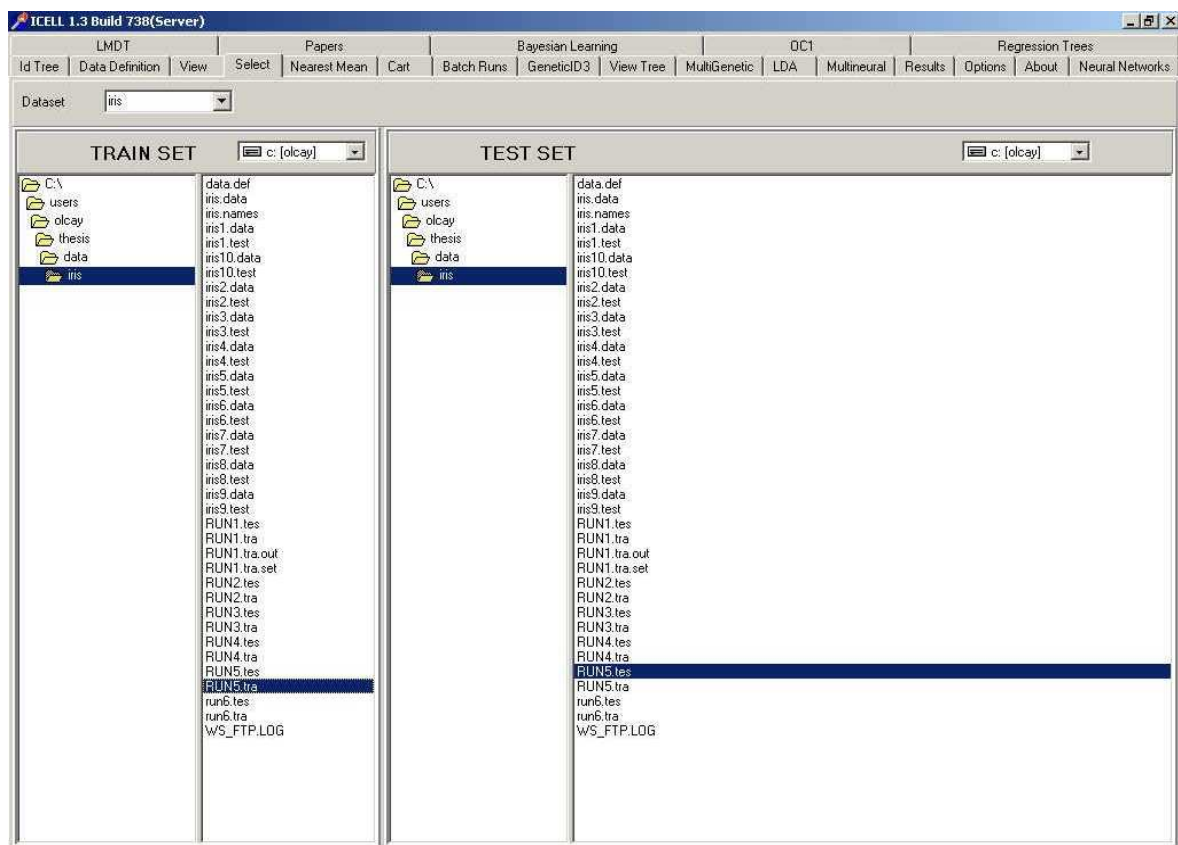


Figure 2.3. Selection of train and test sets for an algorithm in ICELL

After running C4.5 algorithm you may view:

- Tree created in the treeview (Figure 2.5).
- Rules generated by this tree in the memo box. Possible options on rules may be obtained by right clicking on the memo box (Figure 2.6).
 - Save rules
 - Display rules as prolog predicates or usual if-then rules.
- The performance of this tree on the test set.
- Training time.
- Current memory usage.
- Tree created as a picture by pressing fifth button on the IdTree tab (Figure 2.7).

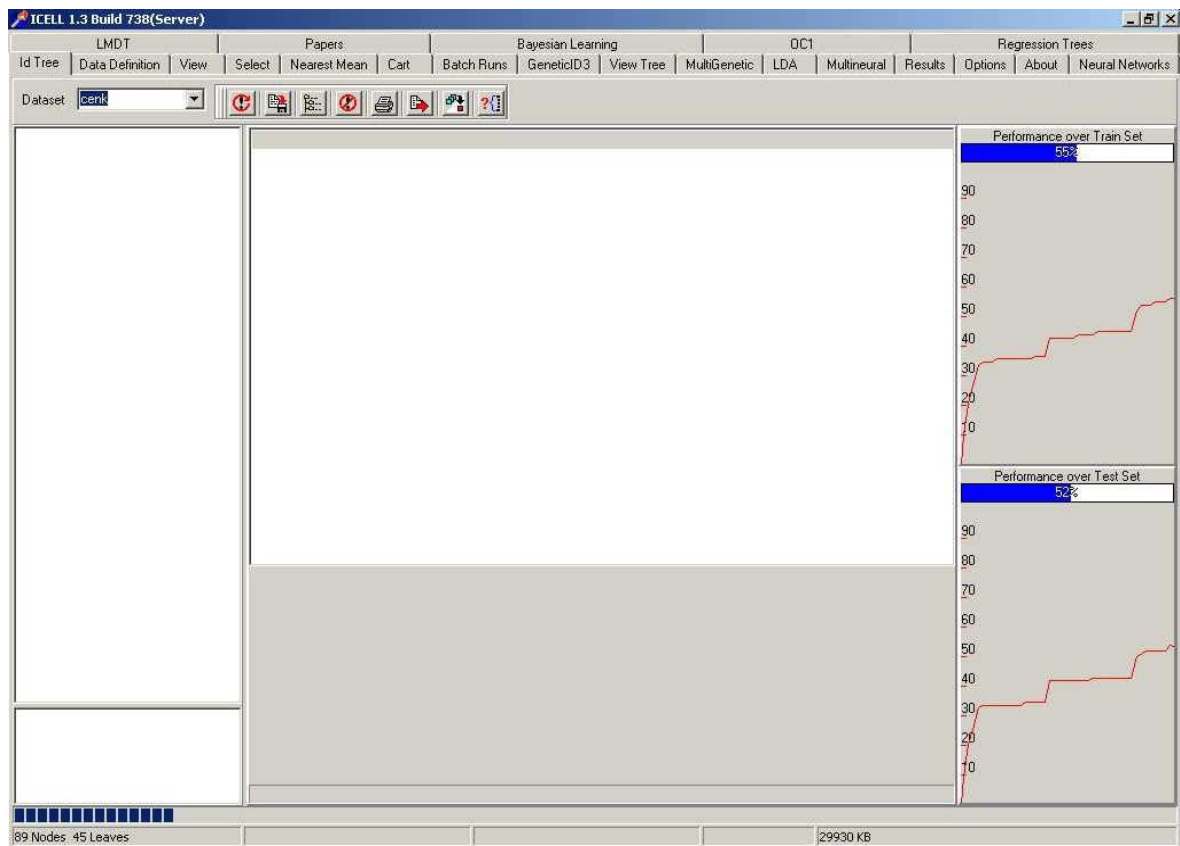


Figure 2.4. Viewing current performance for an algorithm in ICELL

You may also view train data reaching each node by pressing on that node in the treeview (Figure 2.8).

By right clicking on the tree view you get another popupmenu (Figure 2.9), by which you may:

- Collapse the tree in the treeview
- Expand the tree in the treeview
- View the tree as a picture
- Save the tree as an id3 file.
- Copy the tree to the Clipboard.
- Create C and Pascal codes for using this tree in another program.

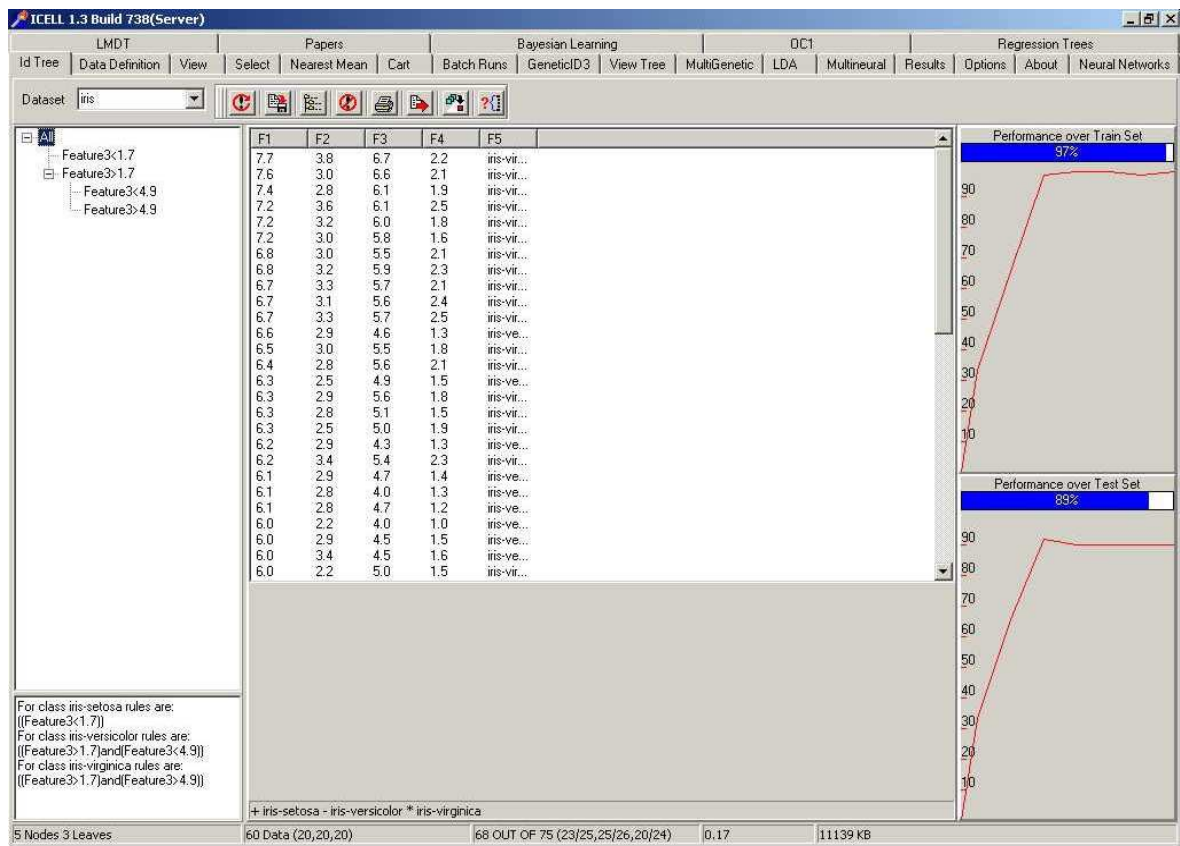


Figure 2.5. Viewing the tree generated with an algorithm in ICELL

To view test and prune data reaching to that node, right click on the data portion of the Idtree tab and select appropriate data set type from the Popupmenu (Figure 2.10).

You may also save the data as a comma separated file by pressing Save data button on the Popupmenu.

For Data visualization purposes ICELL also supports viewing the data on two dimensions (Features of the dataset). To accomplish this:

- Right Click on the panel under the data portion (Figure 3.1).
- Select Set-X-Axis Feature and write the feature number that will be displayed on the X-axis.
- Select Set-Y-Axis Feature and write the feature number that will be displayed on

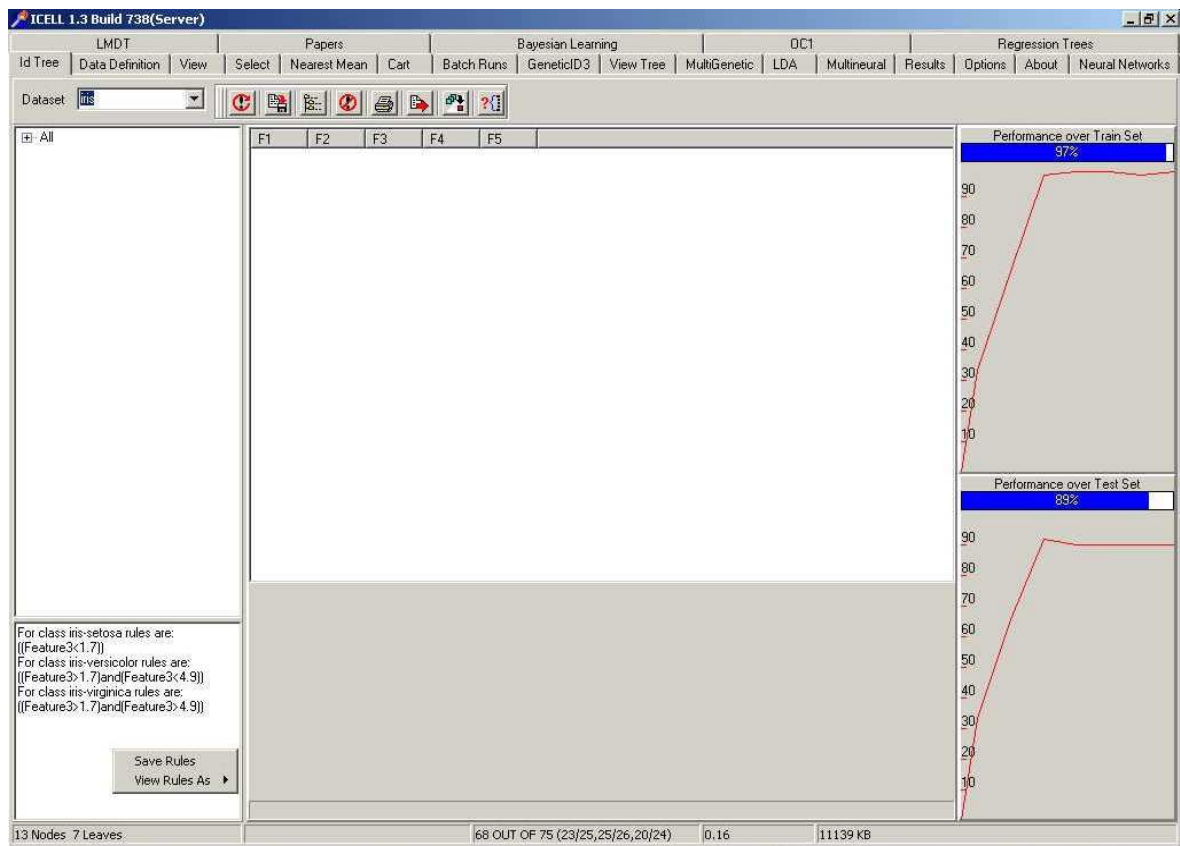


Figure 2.6. Viewing the options for rules generated with an algorithm in ICELL

the Y-axis.

- If you want to save the picture select one of the Save Picture as Bmp or Save Picture as Eps options.

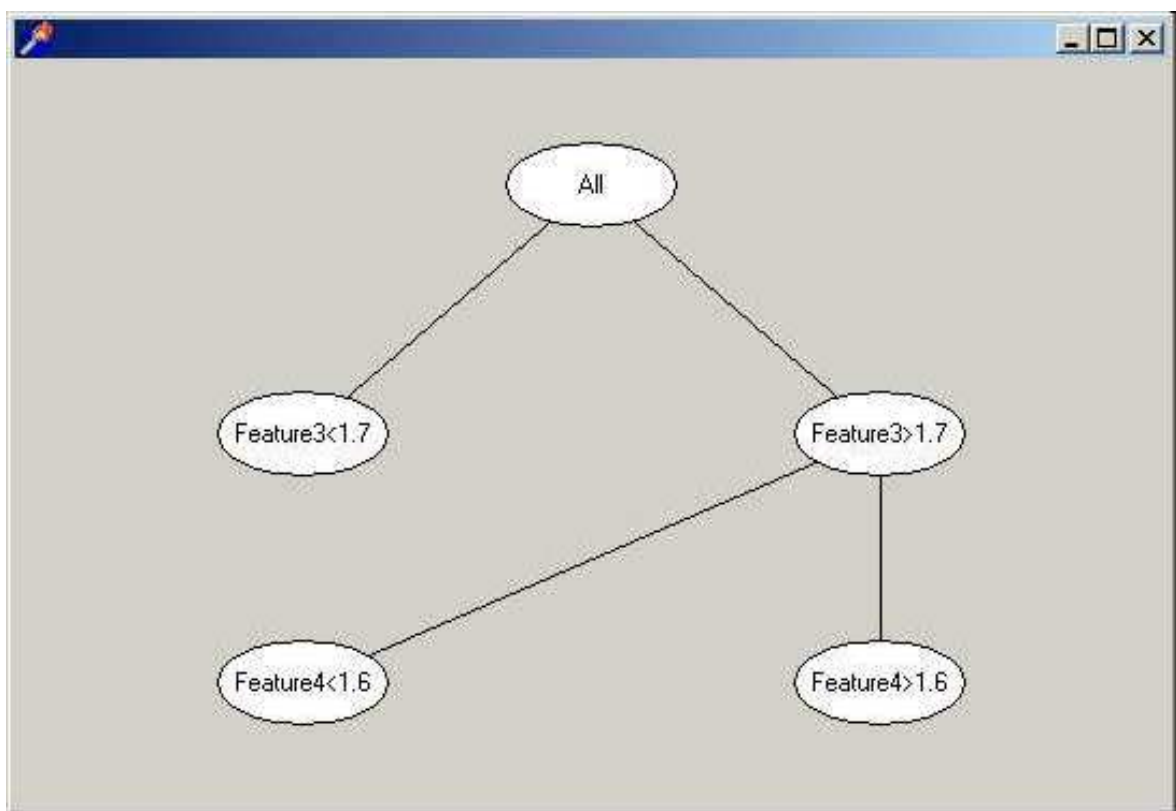


Figure 2.7. Viewing tree as a picture that is generated with an algorithm in ICELL

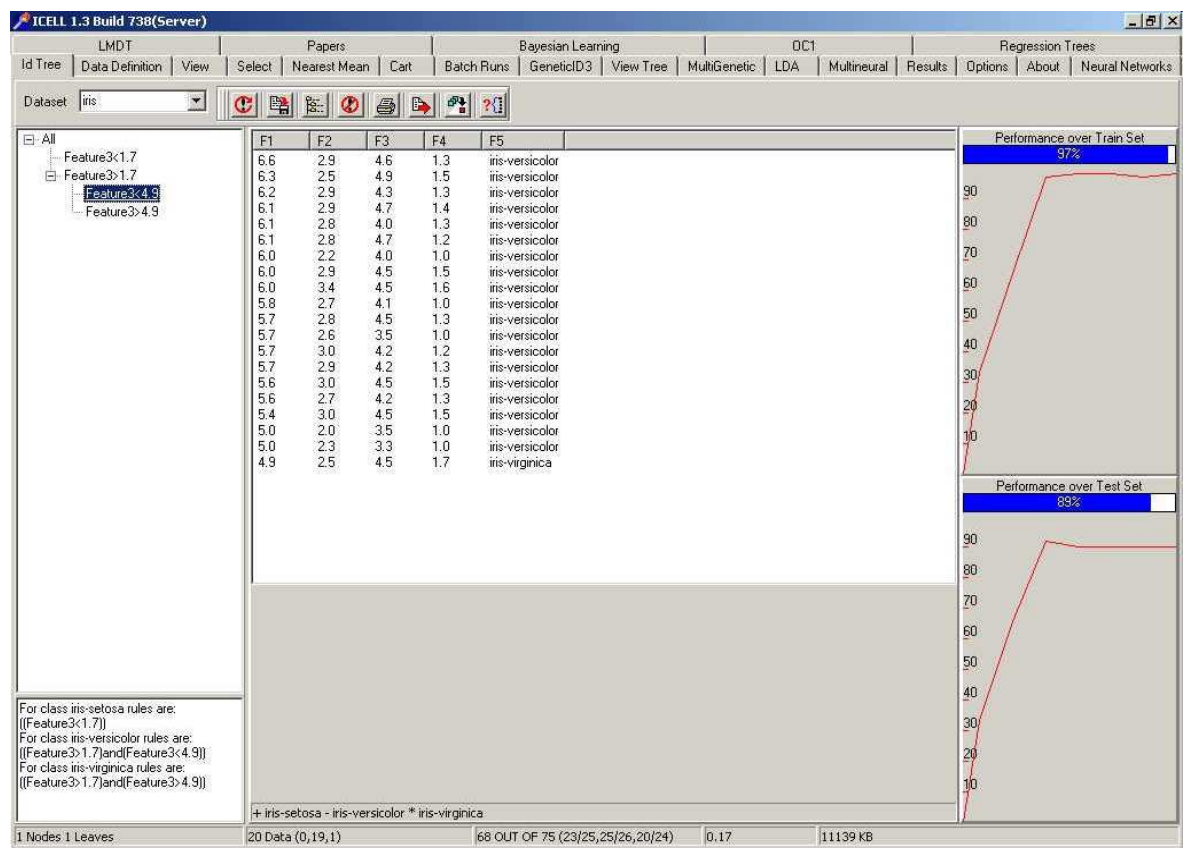


Figure 2.8. Viewing the data on a decision tree node in ICELL

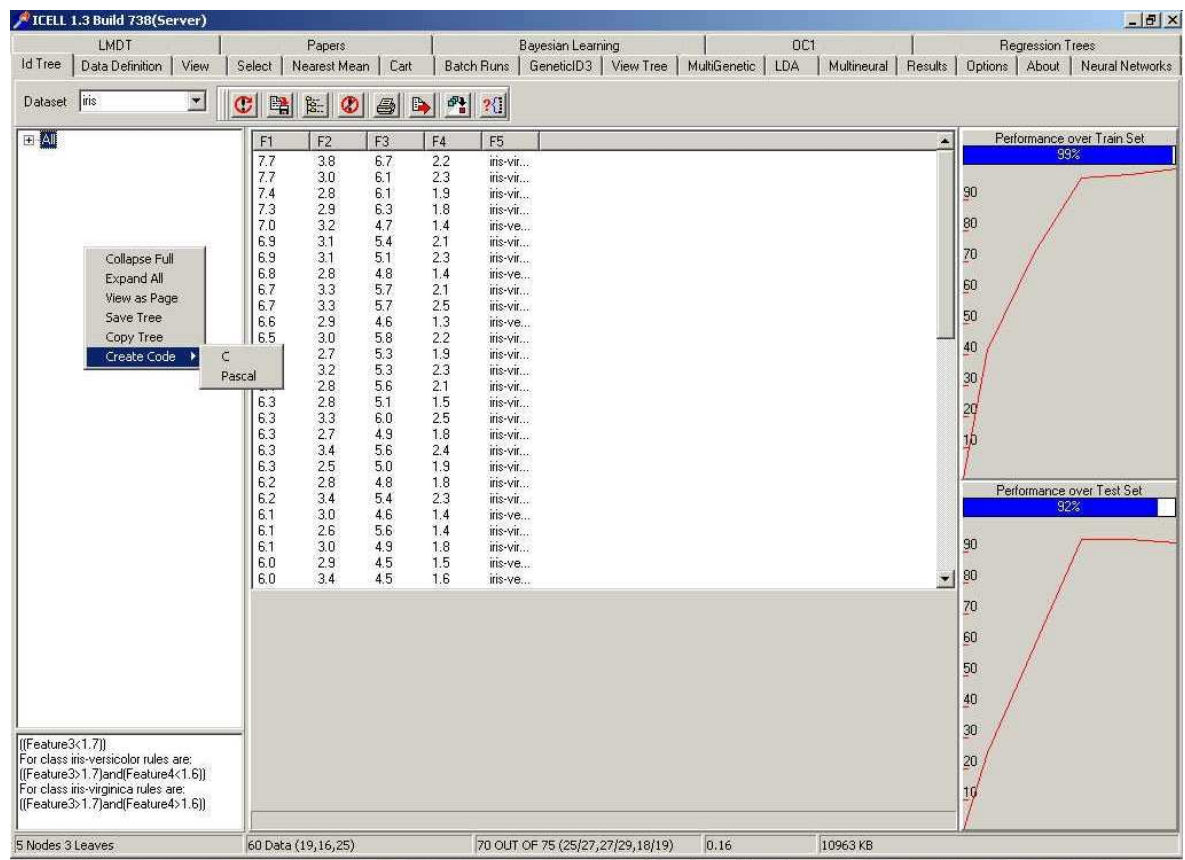


Figure 2.9. The popupmenu for the decision tree in ICELL

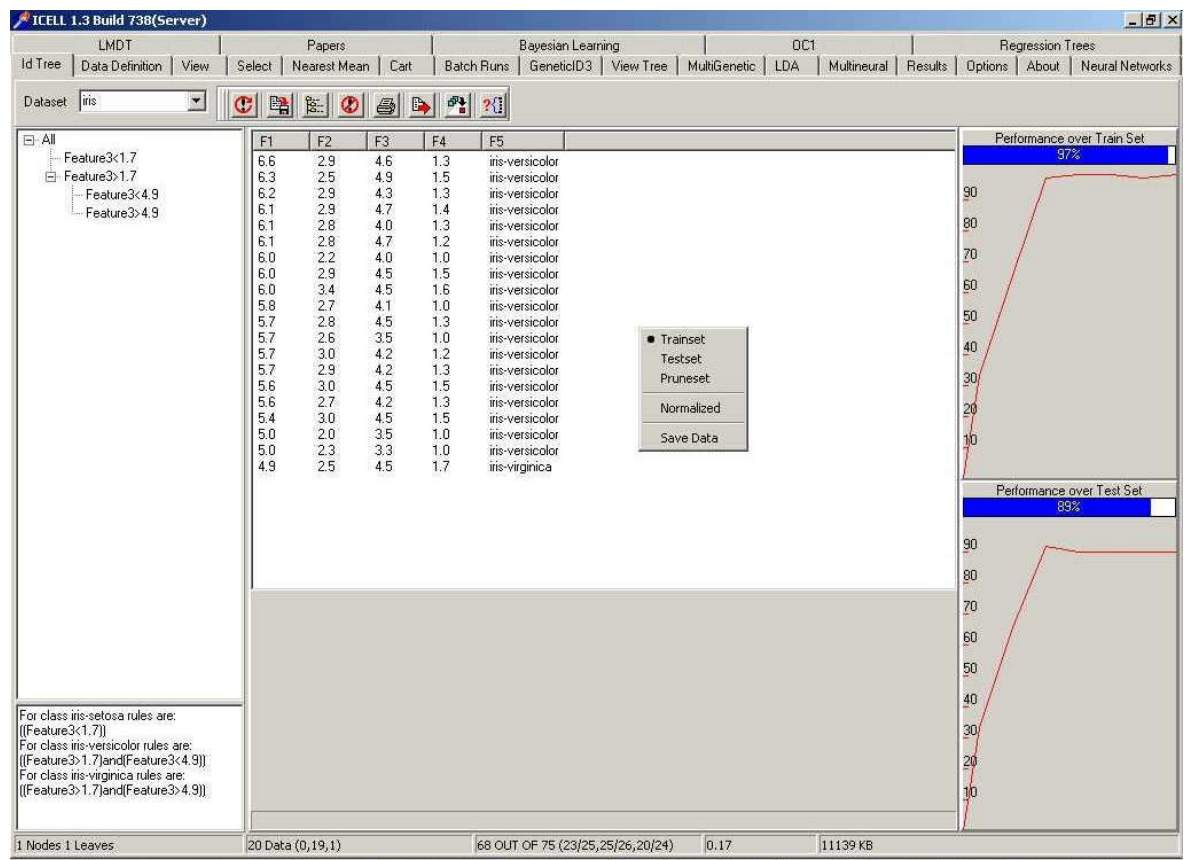


Figure 2.10. The popuption menu for the data portion in ICELL

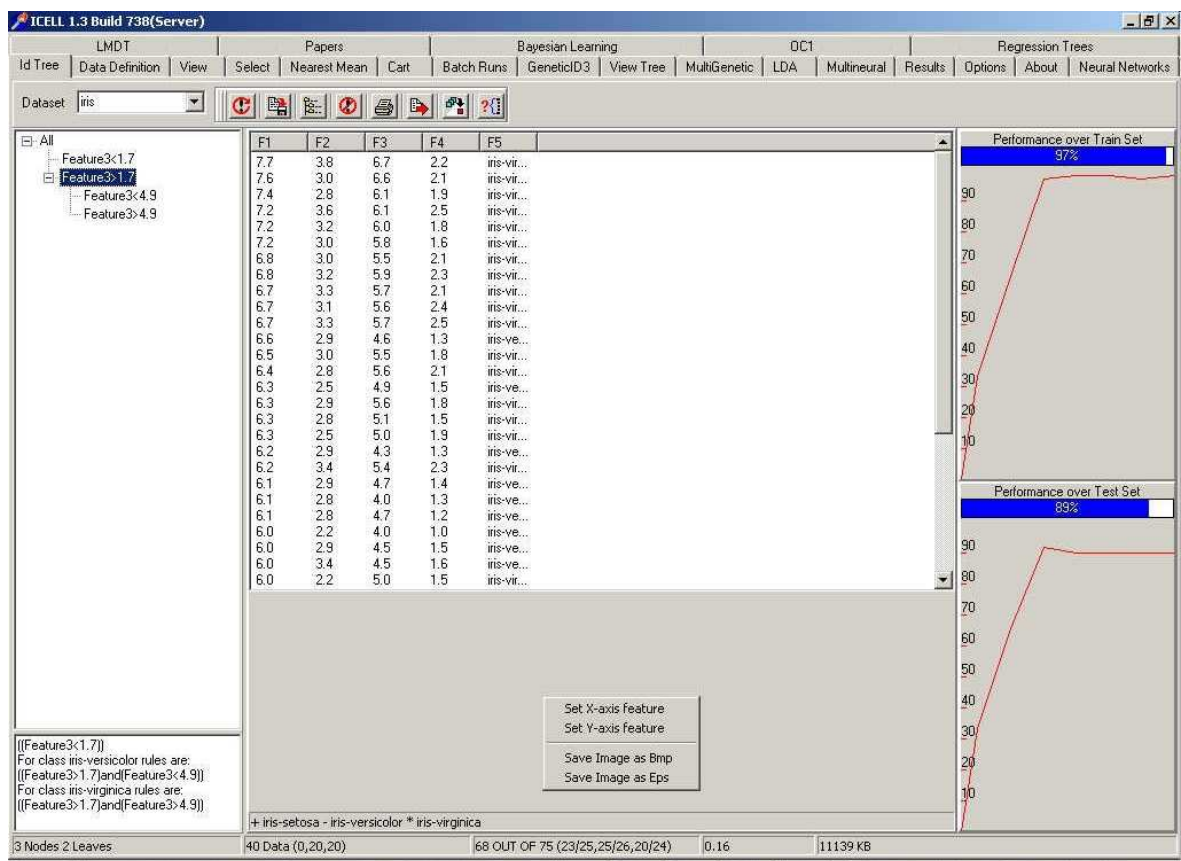


Figure 2.11. The popupmenu for the data visualization portion in ICELL

3. Nearest Mean Classifier

You can run Nearest Mean algorithm from ICELL in two ways:

- ICELL will randomly divide your dataset into two parts (train set, test set), train Nearest Mean algorithm on the train set and test the resulting tree on the test set. To accomplish this:

- Select one of the possible datasets from the dataset combobox.

After running the algorithm you can save train and test files by pressing second button on the Nearest Mean tab.

- You will select the train and test sets and ICELL will run C4.5 algorithm again on your selected train set and test the resulting tree on your test set. To accomplish this:

- Select one of the possible datasets from the dataset combobox on the select tab.
 - Select training set from the train set panel, and test set from the test set panel.
 - Press third button on the IdTree tab.

After running the algorithm, you may validate the algorithm using the test set as train set and train set as test set. For this press first button on the Nearest Mean tab.

While running Nearest Mean algorithm you may:

- Stop running the algorithm by pressing fourth button on the Nearest Mean tab.

After running Nearest Mean algorithm you may:

- View the data on each class by pressing appropriate class on the listview (Figure 3.1).

ICELL 1.3 Build 738(Server)

Dataset: iris

	F1	F2	F3	F4	F5
iris-setosa	6.8	2.8	4.8	1.4	iris-ve...
iris-versicolor	6.6	2.9	4.6	1.3	iris-ve...
iris-versicolor	6.5	2.8	4.6	1.5	iris-ve...
iris-versicolor	6.3	2.5	4.9	1.5	iris-ve...
iris-versicolor	6.2	2.9	4.3	1.3	iris-ve...
iris-versicolor	6.1	2.9	4.7	1.4	iris-ve...
iris-versicolor	6.1	2.8	4.0	1.3	iris-ve...
iris-versicolor	6.1	2.8	4.7	1.2	iris-ve...
iris-versicolor	6.0	2.2	4.0	1.0	iris-ve...
iris-versicolor	6.0	2.9	4.5	1.5	iris-ve...
iris-versicolor	6.0	2.7	5.1	1.6	iris-ve...
iris-versicolor	6.0	3.4	4.5	1.6	iris-ve...
iris-versicolor	5.8	2.7	4.1	1.0	iris-ve...
iris-versicolor	5.8	2.6	4.0	1.2	iris-ve...
iris-versicolor	5.7	2.8	4.5	1.3	iris-ve...
iris-versicolor	5.7	2.6	3.5	1.0	iris-ve...
iris-versicolor	5.7	3.0	4.2	1.2	iris-ve...
iris-versicolor	5.7	2.9	4.2	1.3	iris-ve...
iris-versicolor	5.6	2.9	3.6	1.3	iris-ve...
iris-versicolor	5.6	3.0	4.5	1.5	iris-ve...
iris-versicolor	5.6	2.7	4.2	1.3	iris-ve...
iris-versicolor	5.4	3.0	4.5	1.5	iris-ve...
iris-versicolor	5.0	2.0	3.5	1.0	iris-ve...
iris-versicolor	5.0	2.3	3.3	1.0	iris-ve...
	0.0534...	-0.67...	0.291...	0.154...	

24 DATA 64 OUT OF 75 0.08 0 KB

Figure 3.1. How to view the data in each class in Nearest Mean classifier in ICELL

4. Classification and Regression Trees

You can run CART algorithm from ICELL in two ways:

- ICELL will randomly divide your dataset into two parts (train set, test set), train CART algorithm on the train set and test the resulting tree on the test set. To accomplish this:

- Select one of the possible datasets from the dataset combobox.

After running the algorithm you can save train and test files by pressing second button on the CART tab.

- You will select the train and test sets and ICELL will run CART algorithm again on your selected train set and test the resulting tree on your test set. To accomplish this:

- Select one of the possible datasets from the dataset combobox on the select tab.

- Select training set from the train set panel, and test set from the test set panel.

- Press third button on the CART tab.

After running the algorithm, you may validate the algorithm using the test set as train set and train set as test set. For this press first button on the CART tab.

The parameters of the CART algorithm are:

- Feature Selection: Two options are available. None and Cart. If none, no feature selection is applied, if CART feature selection is applied (Figure 4.1).
- Epsilon: Epsilon parameter used in CART.

You may stop running the CART algorithm by pressing fourth button on the CART tab.

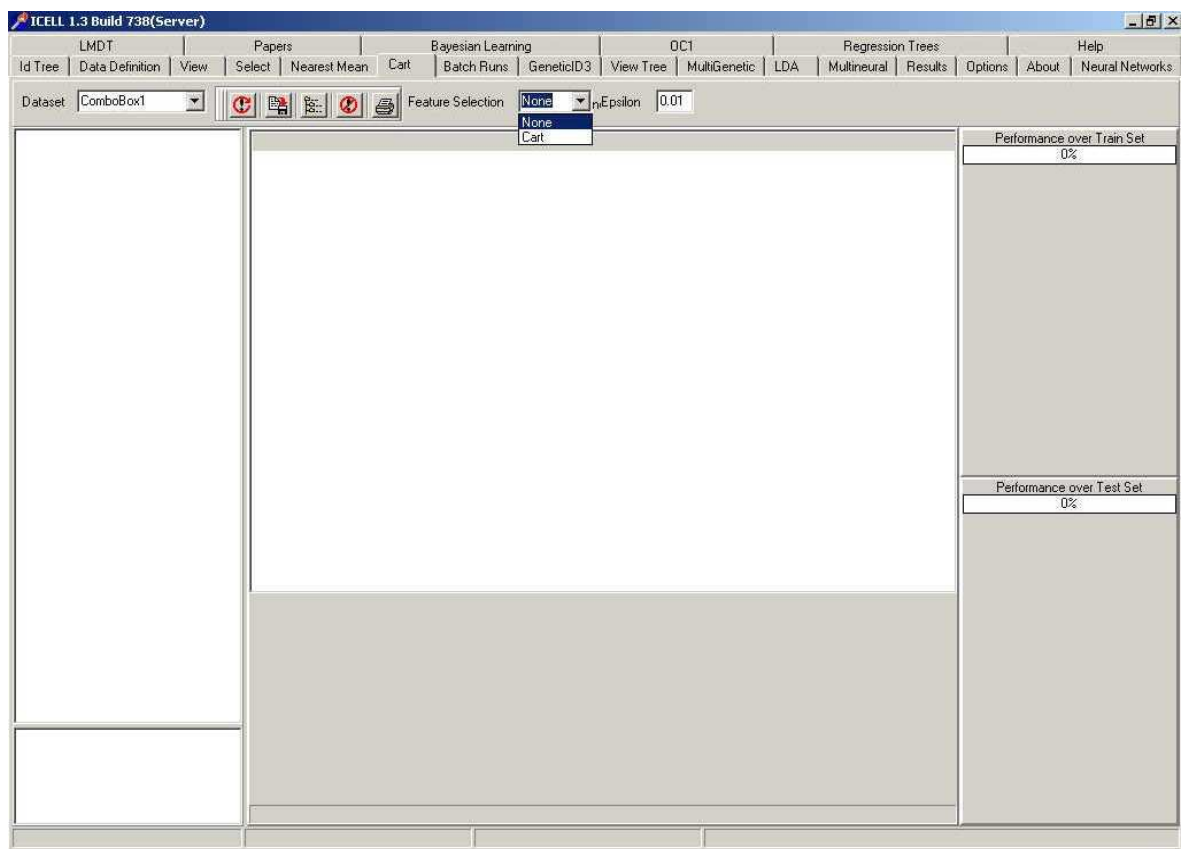


Figure 4.1. Feature Selection option in CART classifier in ICELL

5. Linear Discriminant Trees & QUEST

You can run LDT algorithm from ICELL in two ways:

- ICELL will randomly divide your dataset into two parts (train set, test set), train LDT algorithm on the train set and test the resulting tree on the test set. To accomplish this:

- Select one of the possible datasets from the dataset combobox.

After running the algorithm you can save train and test files by pressing second button on the LDA tab.

- You will select the train and test sets and ICELL will run LDT algorithm again on your selected train set and test the resulting tree on your test set. To accomplish this:

- Select one of the possible datasets from the dataset combobox on the select tab.
- Select training set from the train set panel, and test set from the test set panel.
- Press third button on the LDA tab.

After running the algorithm, you may validate the algorithm using the test set as train set and train set as test set. For this press first button on the LDA tab.

There are several parameters of the LDT algorithm. These are:

- Class Division: Since LDT is a two-class classifier, if a dataset has more than two classes, these classes must be divided into two class groups. This class separation could be done via (Figure 5.1)
 - Exchange Method
 - Selection Method
 - Exchange Method starting with two mostly differing classes
 - Selection Method starting with two mostly differing classes
 - Exhaustive Search for the best class separation

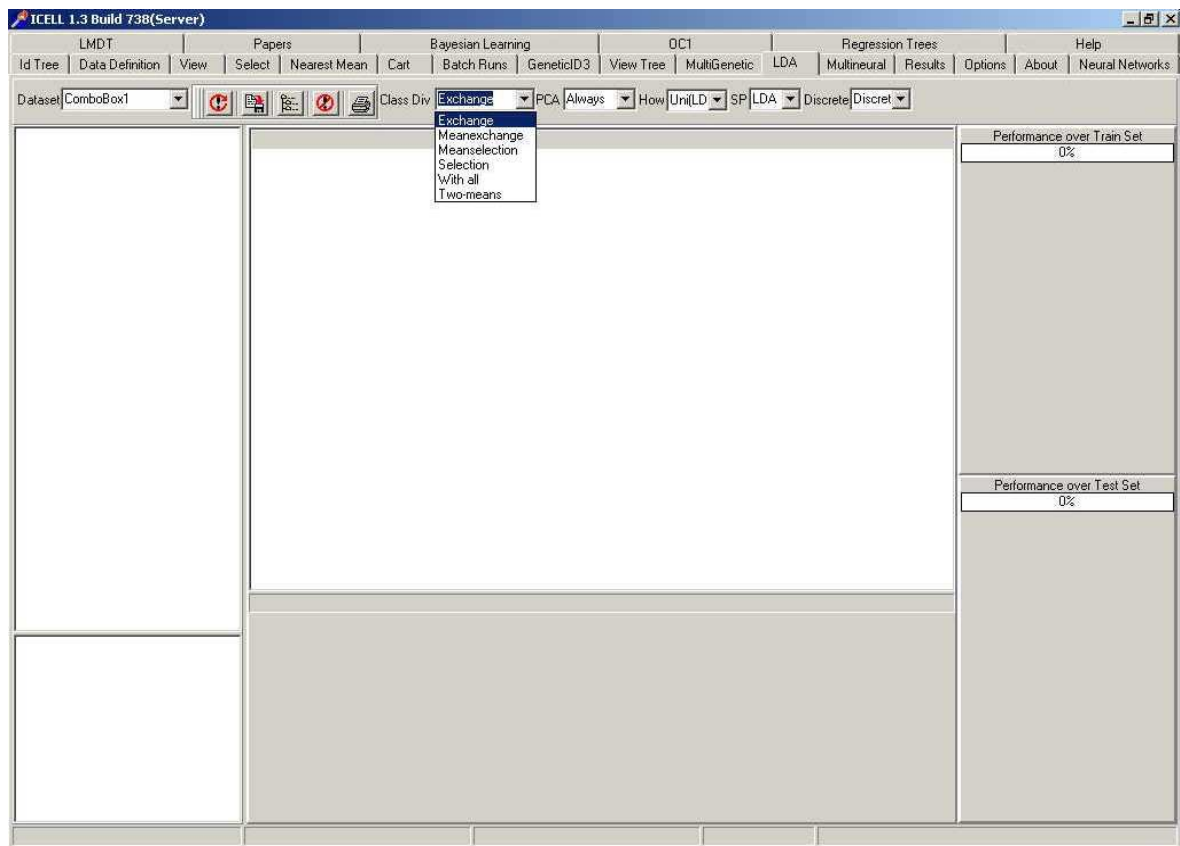


Figure 5.1. How to set the class separation option in LDT classifier in ICELL

- Two-means clustering
- PCA: If the covariance matrix is singular, w could not be calculated. To avoid this, feature extraction is done using PCA. If this option is always, LDT always uses PCA to reduce dimensions. If it is when required, PCA is done when S is singular (Figure 5.2).
- How: LDT can produce three types of decision trees (Figure 5.3).
 - Decision tree with only univariate nodes everywhere
 - Decision tree with only multivariate linear nodes everywhere
 - Decision tree univariate nodes or multivariate linear nodes everywhere.

The type of the decision tree is selected using this How parameter. This parameter take the values

- Uni (LDA) decision tree with only univariate nodes, the splits are calculated as in LDT
- Uni (QDA) decision tree with only univariate nodes, the splits are calculated

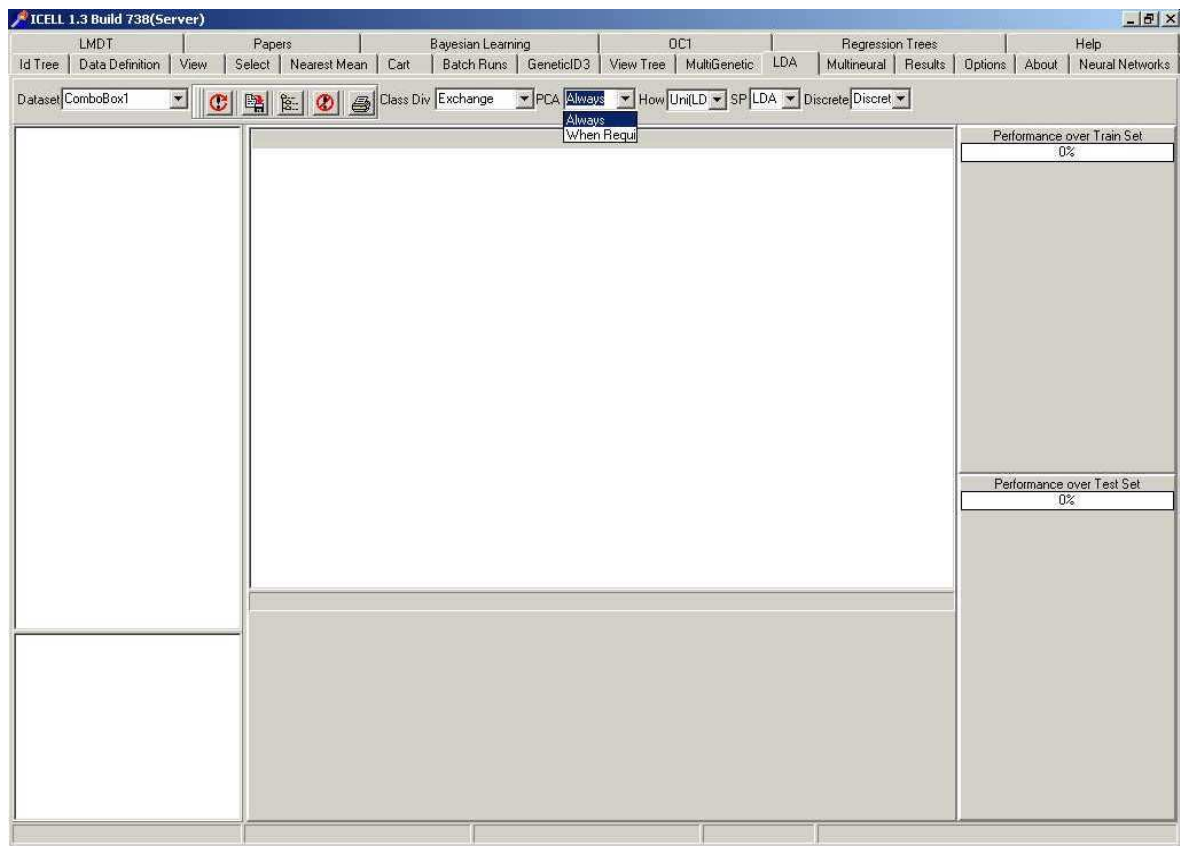


Figure 5.2. How to set the pca option in LDT classifier in ICELL

as in QUEST

- Multi (LDA) decision tree with only multivariate nodes, the splits are calculated as in LDT
- Multi (QDA) decision tree with only multivariate nodes, the splits are calculated as in QUEST
- Uni-Multi (5x2) decision tree with univariate or multivariate nodes, 5x2 cv f test decides type of node
- Uni-Multi (sphericity) decision tree with univariate or multivariate nodes, sphericity tests decides type of node
- SP: The split point in multivariate LDT could be calculated as an exhaustive search or using the result of the equation. This parameter decides the split point's calculation type (Figure 5.4). It is
 - C4.5 in Exhaustive Search
 - LDA in LDT

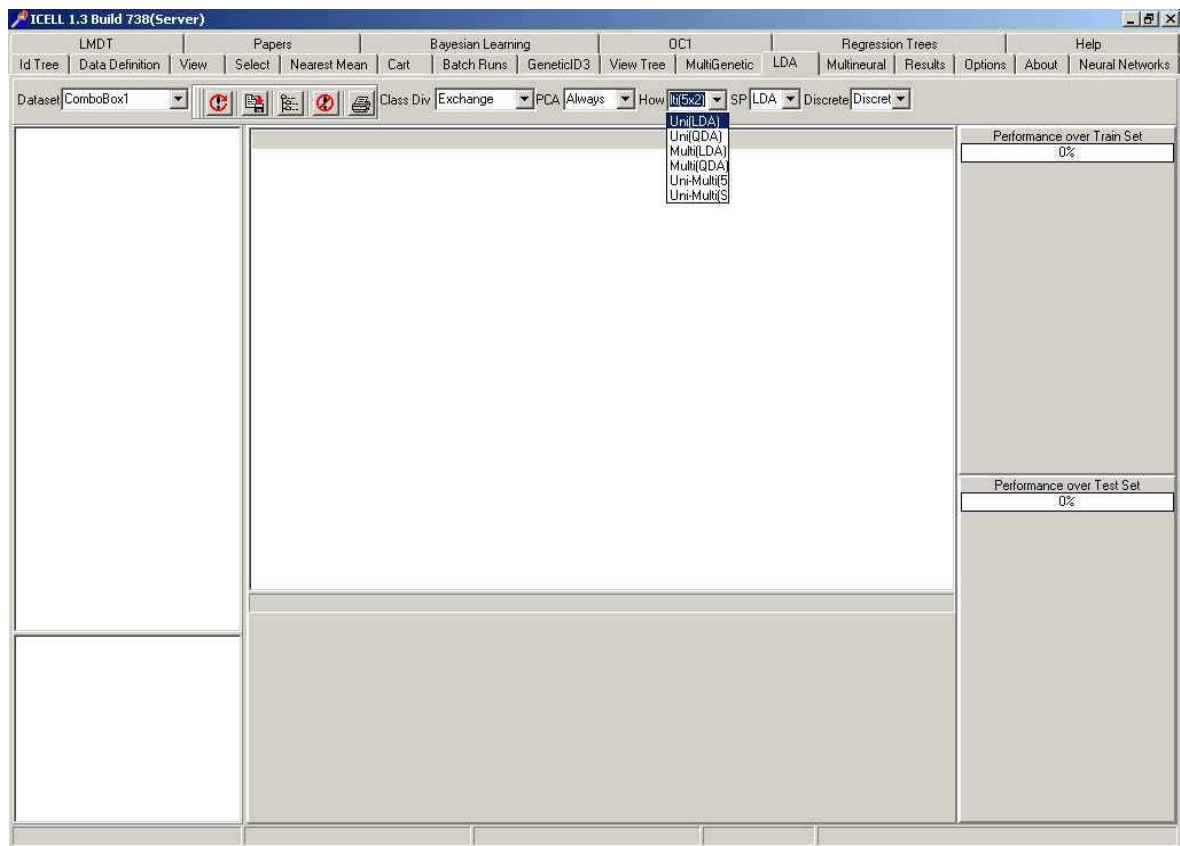


Figure 5.3. How to set the type of decision tree option in LDT classifier in ICELL

- QDA in QUEST
- Discrete: Univariate LDT may use discrete variables as (Figure 5.5)
 - Discrete
 - Real by realizing them
 - Mixed using as discrete and real version of them and selecting best of it

You may stop running the LDT algorithm by pressing fourth button on the LDA tab.

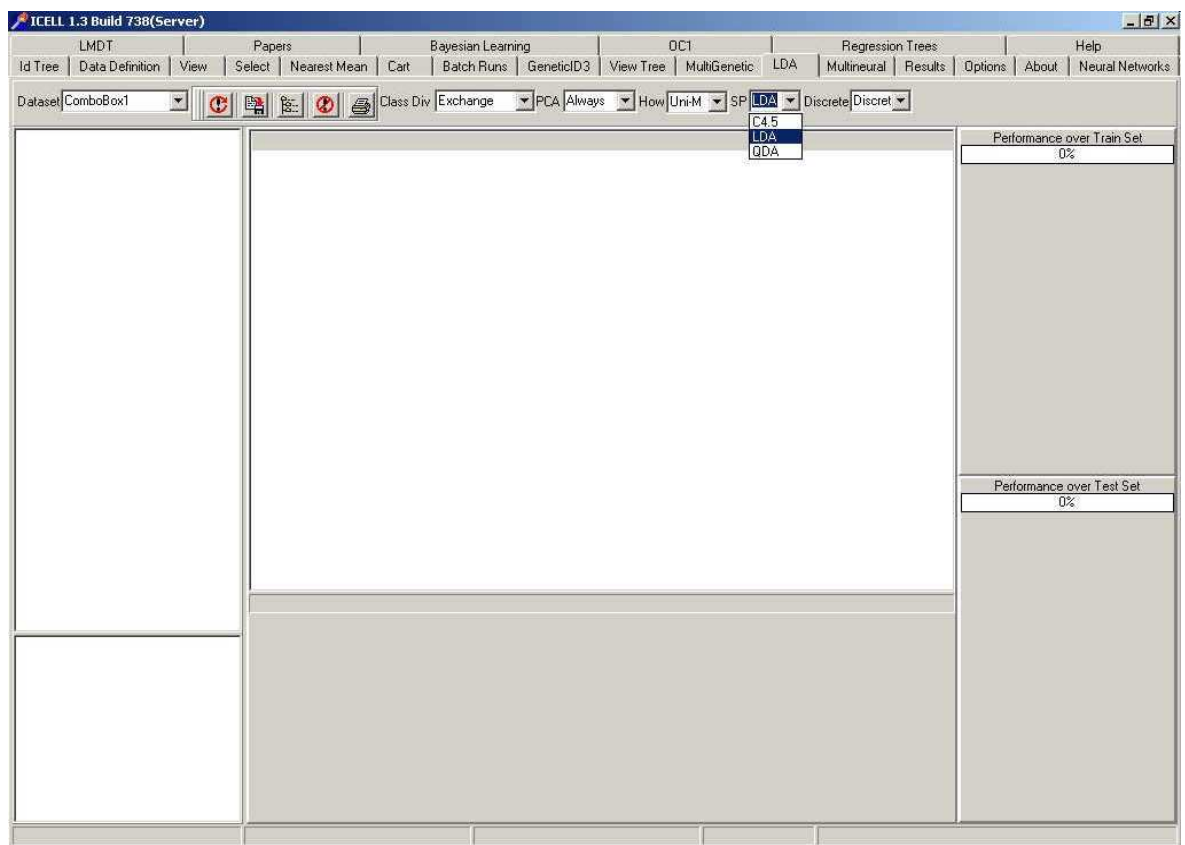


Figure 5.4. How to set the type calculation of split point option in LDT classifier in ICELL

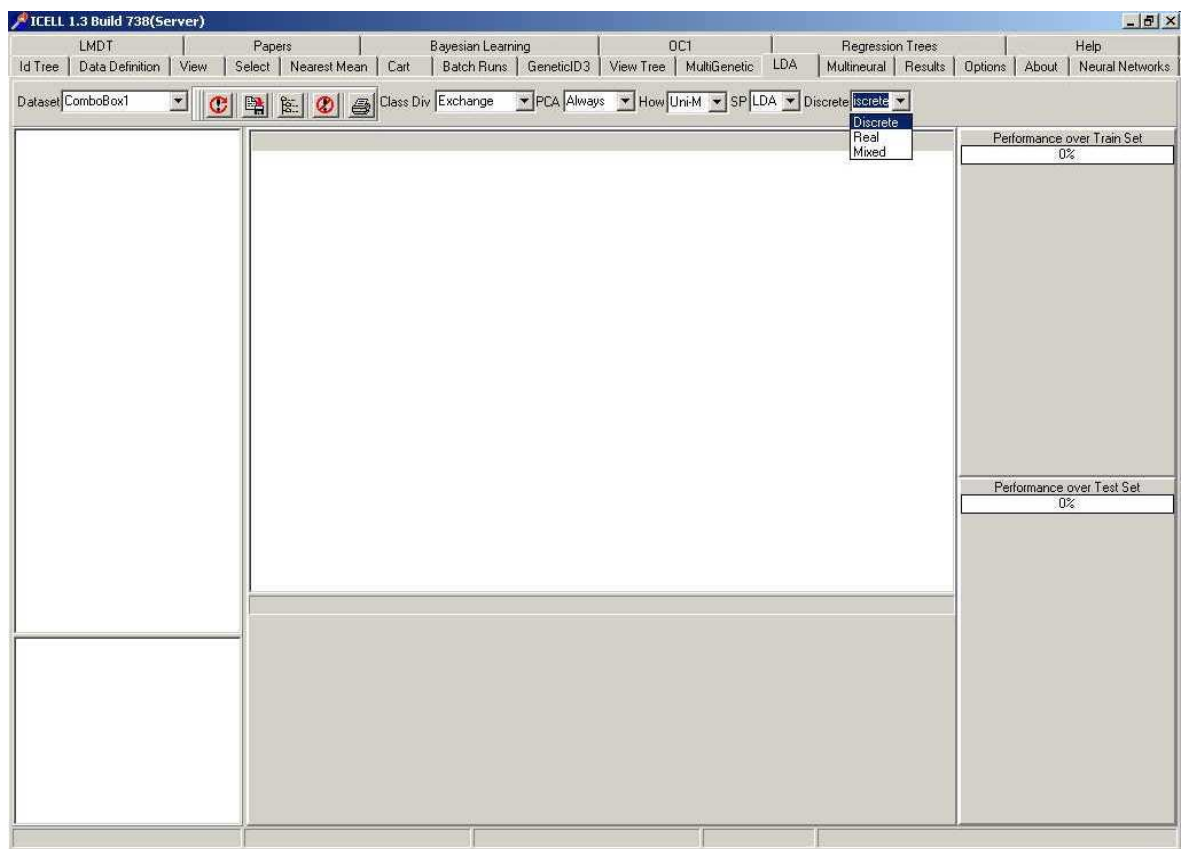


Figure 5.5. How to set the usage of discrete variable in univariate LDT classifier in ICELL

6. ID-LP & ID-MLP & Omnivariate Decision Tree

You can run ODT algorithm from ICELL in two ways:

- ICELL will randomly divide your dataset into two parts (train set, test set), train ODT algorithm on the train set and test the resulting tree on the test set. To accomplish this:

- Select one of the possible datasets from the dataset combobox.

After running the algorithm you can save train and test files by pressing second button on the MultiNeural tab.

- You will select the train and test sets and ICELL will run ODT algorithm again on your selected train set and test the resulting tree on your test set. To accomplish this:

- Select one of the possible datasets from the dataset combobox on the select tab.

- Select training set from the train set panel, and test set from the test set panel.

- Press third button on the MultiNeural tab.

After running the algorithm, you may validate the algorithm using the test set as train set and train set as test set. For this press first button on the MultiNeural tab.

There are several parameters of the ODT algorithm. These are:

- Class Division: Since ODT uses perceptrons as two-class classification problems, if a dataset has more than two classes, these classes must be divided into two class groups. This class separation could be done via (Figure 6.1)
 - Exchange Method
 - Selection Method
 - Exchange Method starting with two mostly differing classes
 - Selection Method starting with two mostly differing classes

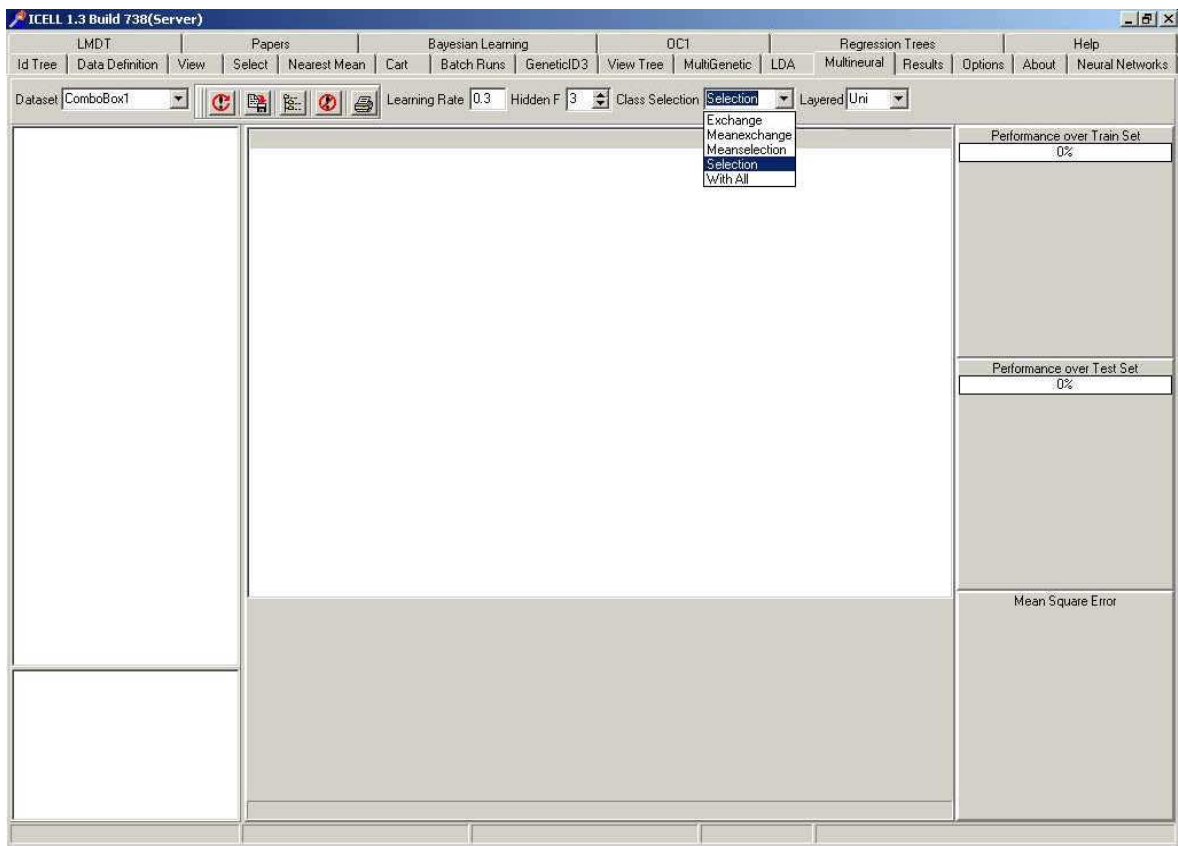


Figure 6.1. How to set the class separation option ODT classifier in ICELL

- Exhaustive Search for the best class separation
- Learning Rate: Learning rate of the perceptron (multilayer or linear) must be set using Learning Rate edit box.
- Hidden F: Finding the correct number of hidden nodes in the multilayer perceptron is not the issue of ODT. Therefore, this hidden factor determines number of hidden nodes in the hidden layers. If hidden factor is 1, number of hidden units in the hidden layer is $(\text{number of inputs} + \text{number of output units}) / 2$. If you increase hidden factor, this number is multiplied with the factor.
- Layered: Complexity of each decision tree node will be determined using this option. Options include (Figure 6.2)
 - Uni: Each node in the decision tree has a linear perceptron. (ID-LP)
 - Multi: Each node in the decision tree has a multilayer perceptron. (ID-MLP)
 - Uni/Multi(t): Type of perceptron is determined using 5x2 cv t test. (ODT)
 - Uni/Multi(f): Type of perceptron is determined using 5x2 cv F test. (ODT)

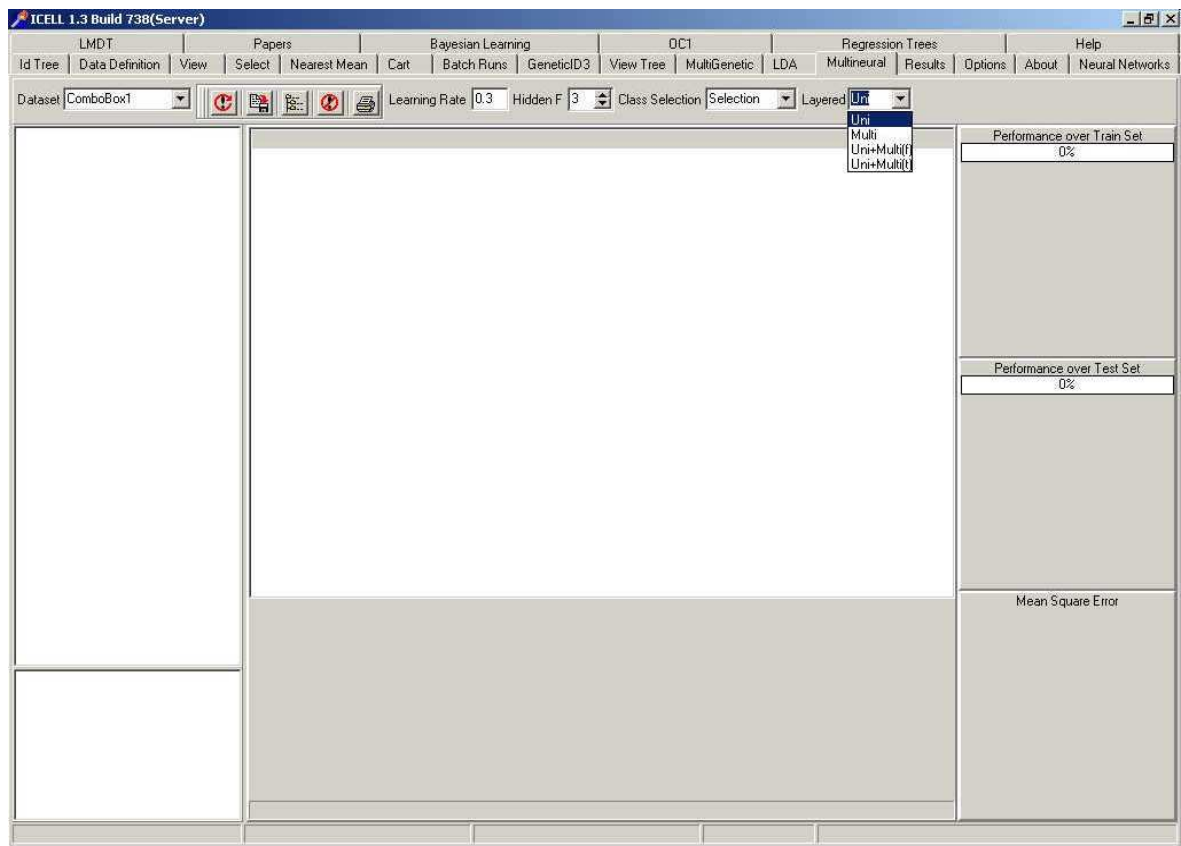


Figure 6.2. How to select type of neural network based decision tree classifier (ID-LP, ID-MLP, ODT) in ICELL

You may stop running the ODT algorithm by pressing fourth button on the Multineural tab.

7. Regression with Multilayer Perceptron

You can run multilayer regression algorithm from ICELL in two ways:

- ICELL will randomly divide your dataset into two parts (train set, test set), train multilayer regression algorithm on the train set and test the resulting tree on the test set. To accomplish this:

- Select one of the possible datasets from the dataset combobox.

After running the algorithm you can save train and test files by pressing second button on the Neural Networks tab.

- You will select the train and test sets and ICELL will run multilayer regression algorithm again on your selected train set and test the resulting tree on your test set. To accomplish this:

- Select one of the possible datasets from the dataset combobox on the select tab.
- Select training set from the train set panel, and test set from the test set panel.
- Press third button on the Neural Networks tab.

After running the algorithm, you may validate the algorithm using the test set as train set and train set as test set. For this press first button on the Neural Networks tab.

The parameters of the multilayer regression algorithm are:

- Momentum: Momentum parameter used in Neural Network training.
- Learning Rate: Learning rate used in Neural Network training.
- Hidden Units: Number of hidden units in the hidden layer.
- Run Size: Number of epochs in Neural Network training.
- Job: Possible regression type. Takes values: (Figure 7.1)
 - Regression
 - RBF-Coop-Regression

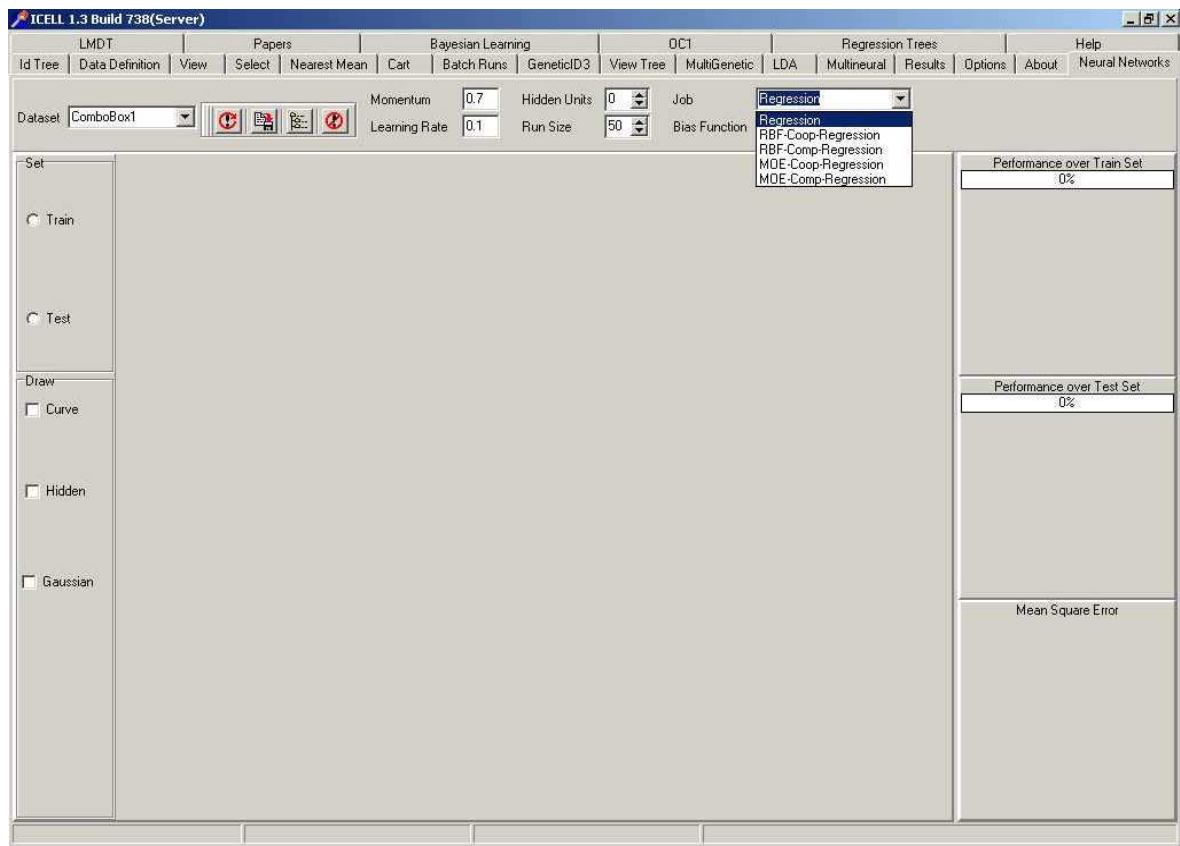


Figure 7.1. How to select type of neural network job in Neural Network Regression in ICELL

- RBF-Comp-Regression
- MOE-Coop-Regression
- MOE-Comp-Regression

You may stop running the multilayer regression algorithm by pressing fourth button on the Neural Networks tab.

8. LMDT

You can run LMDT algorithm from ICELL in two ways:

- ICELL will randomly divide your dataset into two parts (train set, test set), train LMDT algorithm on the train set and test the resulting tree on the test set. To accomplish this:

- Select one of the possible datasets from the dataset combobox.

After running the algorithm you can save train and test files by pressing second button on the LMDT tab.

- You will select the train and test sets and ICELL will run LMDT algorithm again on your selected train set and test the resulting tree on your test set. To accomplish this:

- Select one of the possible datasets from the dataset combobox on the select tab.
 - Select training set from the train set panel, and test set from the test set panel.
 - Press third button on the LMDT tab.

After running the algorithm, you may validate the algorithm using the test set as train set and train set as test set. For this press first button on the LMDT tab.

The parameters of the LMDT algorithm are:

- Feature Selection
- A
- B
- Delta

You may stop running the LMDT algorithm by pressing fourth button on the LMDT tab.

9. OC1

You can run OC1 algorithm from ICELL in two ways:

- ICELL will randomly divide your dataset into two parts (train set, test set), train OC1 algorithm on the train set and test the resulting tree on the test set. To accomplish this:

- Select one of the possible datasets from the dataset combobox.

After running the algorithm you can save train and test files by pressing second button on the OC1 tab.

- You will select the train and test sets and ICELL will run OC1 algorithm again on your selected train set and test the resulting tree on your test set. To accomplish this:

- Select one of the possible datasets from the dataset combobox on the select tab.

- Select training set from the train set panel, and test set from the test set panel.

- Press third button on the OC1 tab.

After running the algorithm, you may validate the algorithm using the test set as train set and train set as test set. For this press first button on the OC1 tab.

The parameters of the OC1 algorithm are:

- Randomrestart: Number of random restarts used in OC1 algorithm
- Local jumps: Number of local jumps used in OC1 algorithm

You may stop running the OC1 algorithm by pressing fourth button on the OC1 tab.

10. Options

10.1. Tree Options

Tree induction options in ICELL could be set in the Options tab under Tree options (Figure 10.1). The options are:

Tree Options

Pruning

☐ Pre-Pruning ☒ Post-Pruning

Partition-Merit Criteria

☒ Information Gain ☐ Gini Index ☐ Square Root

Split No: Overfitting Criteria

Prune Percentage

Figure 10.1. Tree induction options in ICELL

- Pruning Options: Type of pruning used. It may be pre-pruning or post-pruning.
- Partition-merit Criteria: Type of partition merit criteria used in selecting best split. Information Gain, Gini Index or Square Root are possible options.
- Split No: In C4.5, ICELL can also use multiple splits for numeric variables. Example tree generated for Iris dataset is shown here.
- Overfitting Criteria: Used for prepruning. If the ratio of the number of data to the total data reaching to any node is smaller than the overfitting criteria, the tree is not grown more. If pruning is set to prepruning and this number is set to 0, then no pruning is done.
- Prune Percentage: Used for postpruning. Percentage of data used for validation in postpruning is set by this option.

10.2. View Options (Viewing train, test and prune data after run)

After running any algorithm with ICELL, you may also view

- Train set
- Test set
- Prune sets.

To do that, go to View tab. Select which set you want to see from View What, select the algorithm you have run from Method and then press View button. You will see the data in a list view (Figure 10.2).

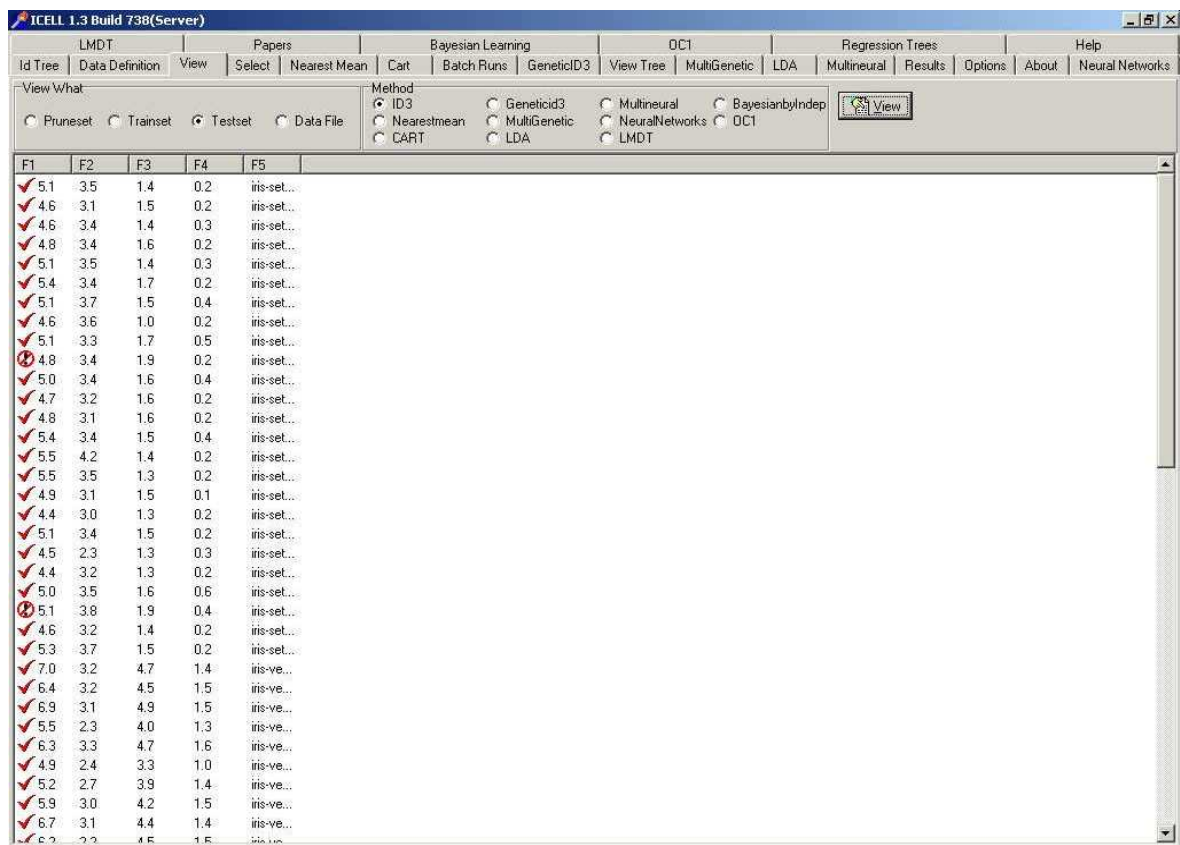


Figure 10.2. The view of train, test or prune data after running the classifier in ICELL

11. Batch Runs

You may also make Batch Runs using ICELL. That means you may run any number of datasets those are defined in ICELL any number of times using any number of algorithms. Each batch operation contain a number of lines, each of them defining a number of runs with a given parameters on a single dataset. Using Add Line button you can add a line of operation to the operations list. To add a single line:

- Press Add Line button.
- Select an algorithm from the dialog box (Figure 11.1).

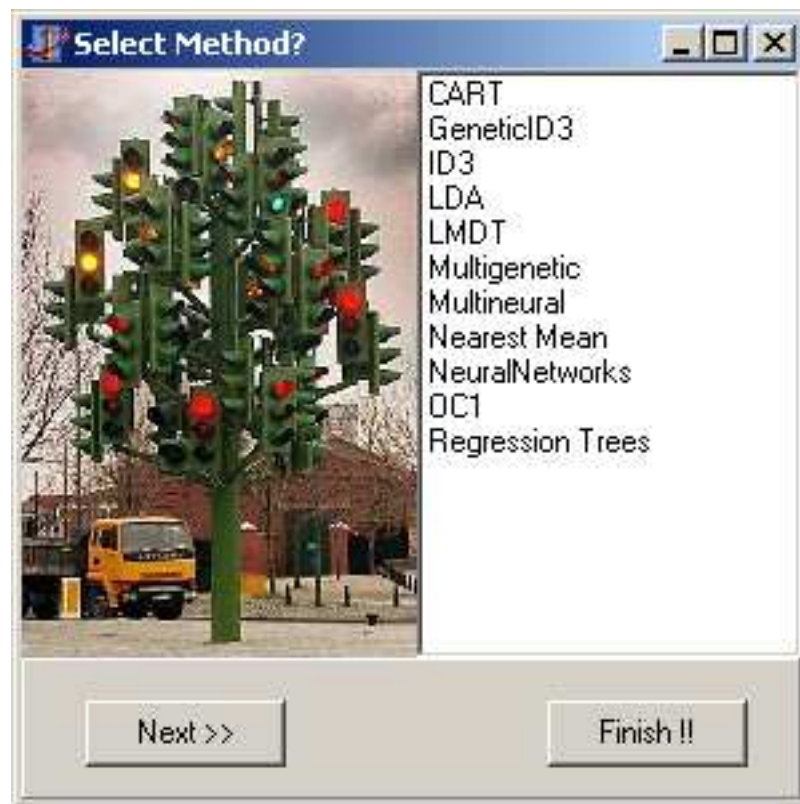


Figure 11.1. Algorithm selection dialog box

- Select a dataset from the dialog box (Figure 11.2).
- Set Tree Induction options from the dialog box (Figure 11.3).
- According to the algorithm you may
 - C4.5: Set the C4.5 algorithm options from the dialog box (Figure 11.4).



Figure 11.2. Dataset selection dialog box

- CART: Set the CART algorithm options from the dialog box (Figure 11.5).
- LDT, QUEST: Set the LDT or QUEST algorithm options from the dialog box (Figure 11.6).
- ID-LP, ID-MLP, ODT: Set the ID-LP, ID-MLP or ODT algorithm options from the dialog box (Figure 11.7).
- MLP-Regression: Set the MLP-Regression options from the dialog box (Figure 11.8).
- LMDT: Set the LMDT algorithm options from the dialog box (Figure 11.9).
- OC1: Set the OC1 algorithm options from the dialog box (Figure 11.10).
- Set the general Batch Run options from the dialog box (Figure 11.11).

After that a line will appear in the Batch list (Figure 11.12). Now you can either add another batch line using the above procedure or run the batch operation using



Figure 11.3. Tree induction options selection dialog box

Run button.

After running the batch list you can see the results of the algorithms in the Results panel (Figure 11.13).

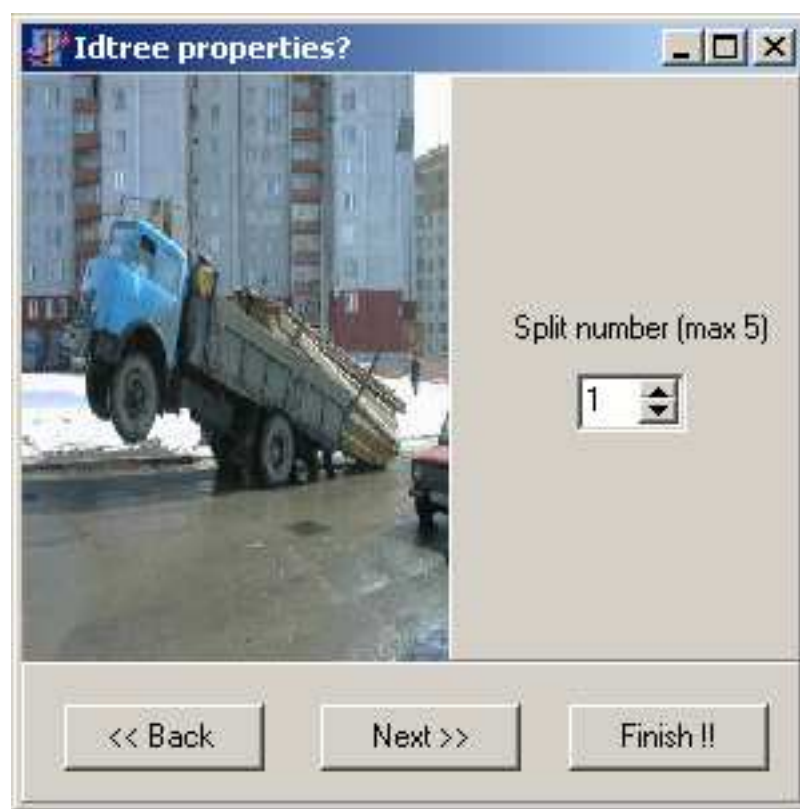


Figure 11.4. C4.5 algorithm parameters selection dialog box



Figure 11.5. CART algorithm parameters selection dialog box



Figure 11.6. LDT, QUEST algorithms parameters selection dialog box



Figure 11.7. ID-LP, ID-MLP, ODT algorithms parameters selection dialog box

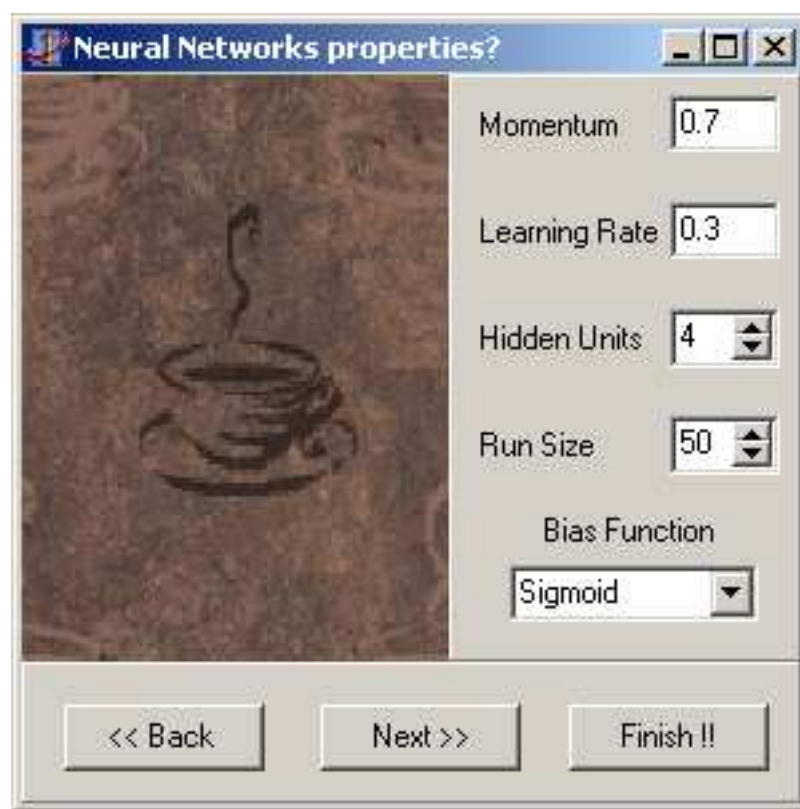


Figure 11.8. MLP-Regression algorithms parameters selection dialog box



Figure 11.9. LMDT algorithm parameters selection dialog box

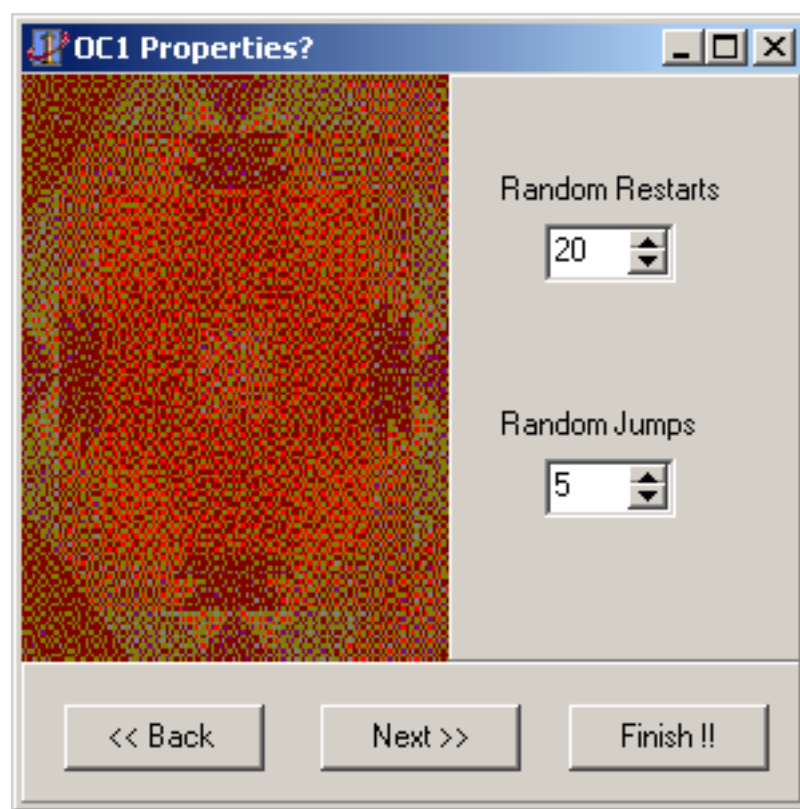


Figure 11.10. OC1 algorithm parameters selection dialog box



Figure 11.11. General batch options selection dialog box

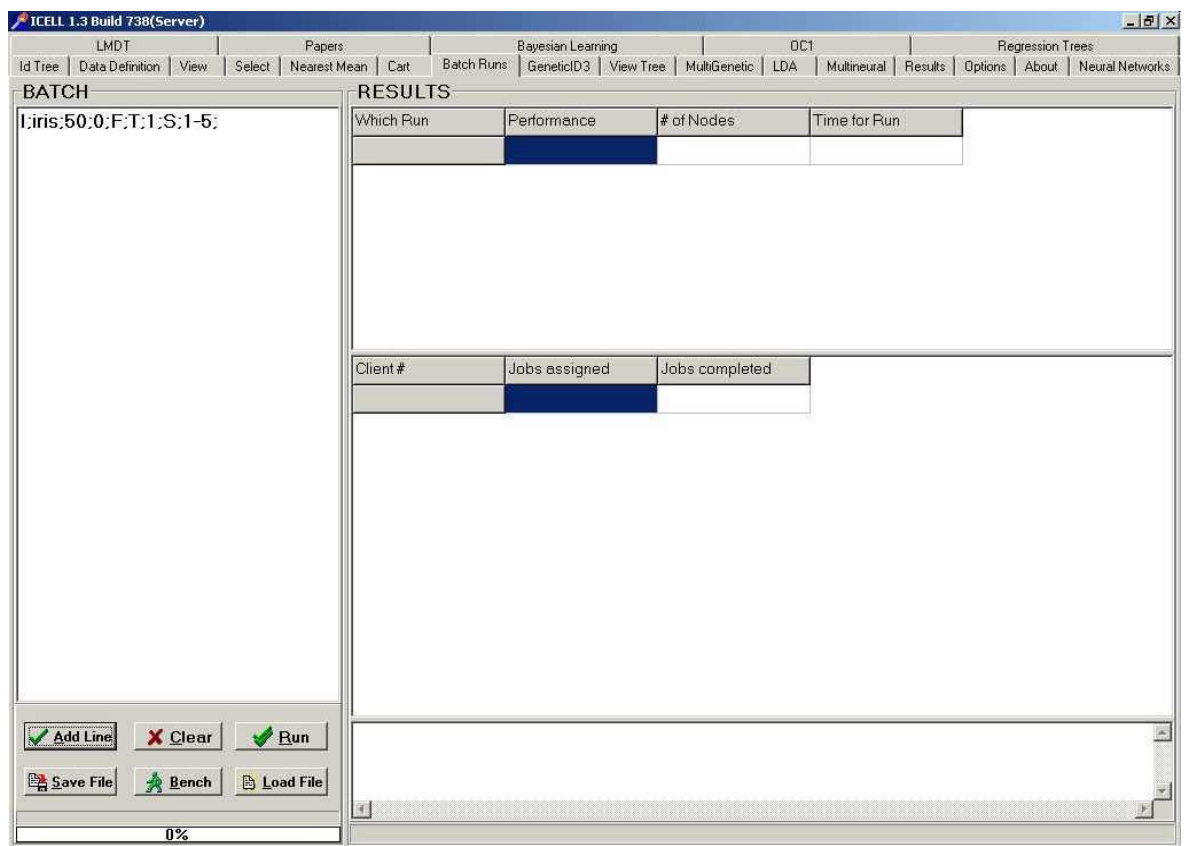


Figure 11.12. Batch run window after adding one line as a batch run in ICELL

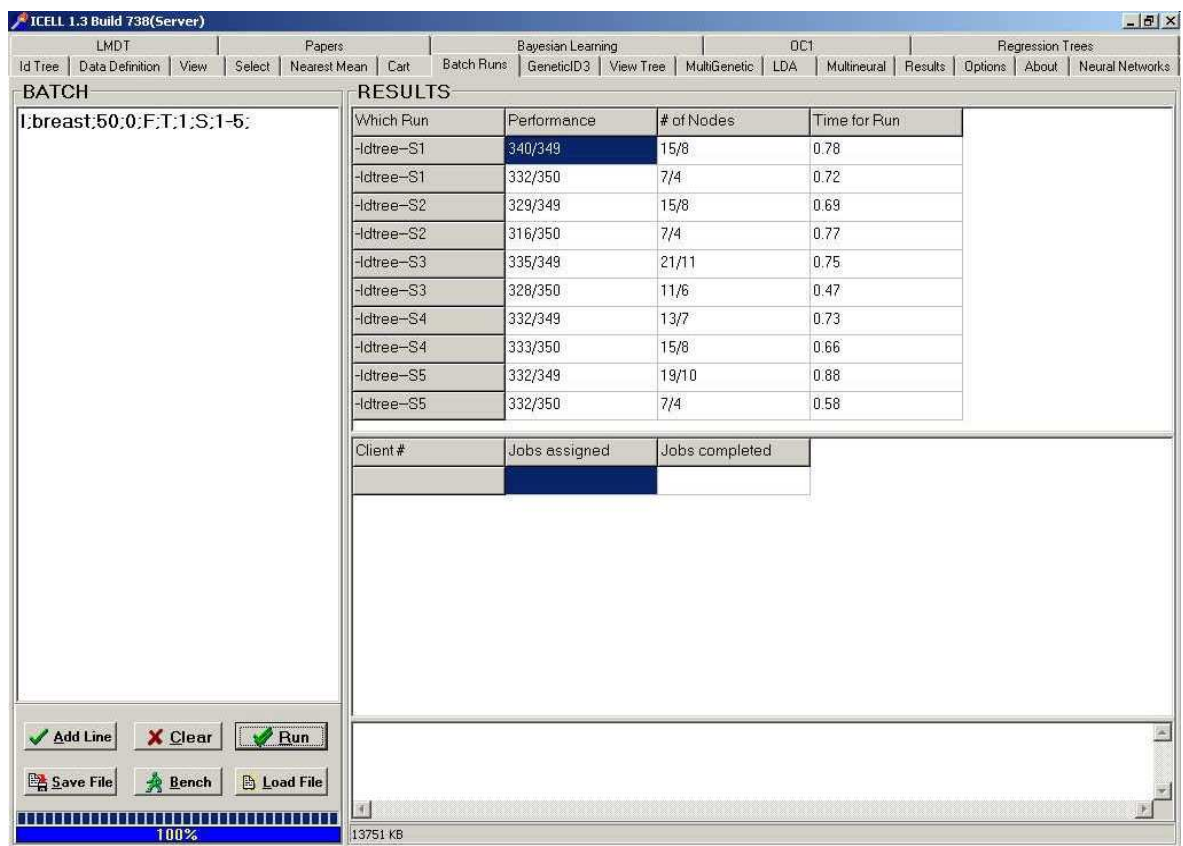


Figure 11.13. Result panel after a batch run in ICELL

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